

Unlocking Strong Supervision: A Data-Centric Study of General-Purpose Audio Pre-Training Methods

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Abstract

Current audio pre-training seeks to learn unified representations for broad audio understanding tasks, but it remains fragmented and is fundamentally bottlenecked by its reliance on weak, noisy, and scale-limited labels. Drawing lessons from vision’s foundational pre-training blueprint, we argue that the audio field must first establish its own large-scale, strong supervision framework. We introduce a new data-centric pipeline that leverages a high-fidelity captioner to create SOTA-quality captions and the first Unified Tag System (UTS) that bridges speech, music, and environmental sounds. We then conduct a systematic comparative study of different pre-training objectives on these strong source data. Our experiments suggest that data quality and coverage are the primary drivers of performance, while the choice of objective dictates downstream task specialization.

1. Introduction

Large-scale pre-training has become the dominant paradigm for developing general-purpose models in computer vision field [48, 58, 65, 68]. In comparison, the audio domain has historically been fragmented, with research siloed into specific sub-fields. This has yielded powerful task-specific models for speech [4, 40, 66], speaker-relevant [19, 72], music [52, 97], and general sounds [11, 13, 33, 47]. While these models excel on respective benchmarks [79, 81, 93], this fragmentation has hindered the development of a single, unified model capable of spanning the full spectrum of audio data.

To achieve true general-purpose audio understanding [74], Audio-Language Pre-training (ALP) has emerged as the most promising strategy. Inspired by the success of Vision-Language Models [44, 50, 65, 78, 94], ALP leverages free-form text as a universal supervisory signal. The prevalent approach is contrastive learning (CLAP) [24, 84],

alongside other methods like generative captioning [77], all aiming to learn representations that are both general-purpose and semantically rich.

However, this paradigm is fundamentally bottlenecked by its supervision source, which lacks both the scale of vision’s multi-billion pair corpora [29, 70] and the quality of a true foundational dataset. Current audio caption datasets barely exceed one million pairs [6, 23, 45, 60] and are often just “LLM-augmented” sets, merely fluent rewrites of existing sparse tags [6, 60]. This reliance on a noisy, semantically-sparse, and scale-limited source fundamentally caps pre-training potential. We argue that instead of mimicking the current web-scale VLM paradigm, the audio domain should first draw inspiration from an earlier, foundational blueprint from vision: the ImageNet paradigm [18]. This approach proved that a large-scale, high-quality, strongly-supervised dataset could unify a field, and was later validated at massive scales via Multi-Tag Classification (MTC) [57, 95]. While PANNs [47] applied MTC in the audio domain, it was still limited by AudioSet [30]’s scope, which fails to truly unify the domains of speech, music, and environmental sounds.

To this end, we introduce a new data-centric paradigm as shown in Fig. 1. We leverage a high-fidelity audio captioner, *Qwen3-Omni-Captioner* [87], to process a diverse audio dataset *CaptionsStew* [2] 400k-subset. From its rich, descriptive captions, we generate two distinct strong supervision sources: (1) a set of high-fidelity SOTA-quality captions for audio-language pre-training, and (2) a large-scale, data-driven Unified Tag System (UTS) for tag-oriented pre-training, which is the first of its kind, with up to 3k tags, that naturally bridges speech, music, and environmental sounds. We then conduct a systematic comparative study of all pre-training objectives. Our results are conclusive: (1) Our UTS-trained models achieve superior out-of-domain generalization (e.g., on speech/music) over the MTC (AudioSet) baseline, despite using 5x less data. (2) Our SOTA-Caps-trained models consistently outperform the [2] baselines across a majority of tasks. This confirms our core hypoth-

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esis: the quality and richness of the supervision source is a primary driver of performance in audio pre-training.

Our contributions are threefold. First, we introduce a novel data-centric pipeline to create the first large-scale Unified Tag System (UTS) that bridges speech, music, and environmental sounds. Second, we conduct a systematic comparative study of different audio pre-training objectives, providing a comprehensive benchmark on this new strong supervision source. Finally, our results suggest that data quality and coverage can be more impactful than data volume, and that the choice of pre-training objective dictates downstream task specialization, offering a clear lesson for the future of audio pre-training.

2. Related Work

2.1. Audio Representation Learning

Audio representation learning aims to develop a single, general-purpose model for diverse understanding tasks. This field has progressed along two parallel tracks: The first is task-specific supervised learning, which has produced high-performing encoders for audio event classification [11, 17, 21, 32, 37, 47], speech recognition systems [66], and speaker recognition models [19, 72]. While these models are strong, their effectiveness is inherently limited by their specific, predefined label sets. The second major approach is self-supervised learning (SSL), which learns from unlabeled data and has yielded powerful, specialized encoders for speech [4, 5, 12, 40], general audio [13, 33, 41, 51], and music [52, 97]. SSL models also show promise for generative tasks such as audio synthesis. While these methods improve generalization within specific domains, the challenge remains to develop a representation that seamlessly integrates speech, music, and environmental sounds while overcoming task-specific limitations.

2.2. Audio-Language Pre-Training

Inspired by the success of Vision-Language Pre-training [44, 50, 65, 78, 94], Audio-Language Pre-training has emerged to learn rich, semantic features by leveraging the natural co-occurrence of audio and text. The field is first dominated by Contrastive Learning [24, 25, 35, 84], with recent extensions have explored combinations with other objectives [61, 62, 89, 96], most notably Generative Pre-training (e.g., Captioning [31, 39, 77]). This has also spurred an evolution in datasets. Influenced by the prevalence of using LLMs as caption re-writers in the vision field [27], this trend has moved from small, human-annotated corpora [1, 23, 45] to large-scale, “LLM-augmented” collections [6, 14, 60, 76] and domain-specific resources [22, 69]. However, most existing datasets, even those “LLM-augmented” are typically derived from existing sparse tags or noisy web-captions; the LLM is

often used merely to rewrite these weak sources into fluent prose. Thus, the semantic richness of the supervision is still constrained by the low-quality original source.

2.3. Vision’s Blueprint for Supervision

The methodologies from vision-based large-scale representation learning provide a clear blueprint for the audio domain. While recent Vision-Language Models (VLMs) have demonstrated the power of web-scale supervision, two key lessons from earlier vision work are particularly relevant. First, the ImageNet [18] paradigm highlighted the transformative effect of a single, large-scale, strongly-supervised dataset. For nearly a decade, pre-training on ImageNet’s 1000-class vocabulary was a dominant transfer learning strategy. More recent models, such as those in [36], further established that a semantically-rich, classification-based dataset can be a powerful foundation for downstream tasks. Second, Multi-Tag Classification (MTC) [95] proved to be an effective and scalable pre-training objective. [57] demonstrated the power of training on billions of Instagram images and their noisy user-generated hashtags, showing that MTC can be a valuable source of strong supervision. Recent VLM [15, 42, 43, 53] show that MTC offers a more direct and structured semantic supervision signal and leads to stronger downstream performance in tasks such as image captioning and retrieval. Our work builds on these vision strategies by applying them to the audio domain. Specifically, we aim to (1) establish a unified foundational audio classification dataset and (2) leverage MTC as the primary form of strong supervision for general audio.

3. Method

3.1. Dataset Construction

We utilize audio data from the *CaptionStew* 400K-subset introduced in [2], which encompasses a diverse range of source including **speech, music, and environmental sounds**. We then employ the *Qwen3-Omni-Captioner* [87], f_{cap} , to generate a detailed natural language caption $c_i = f_{cap}(a_i)$ for each audio clip a_i . These captions are then analyzed by a large language model (LLM), which extracts relevant tags based on the semantic content. This process results in a comprehensive and diverse tag vocabulary derived from the entire audio-caption dataset. Details of the data source can be found in Supplementary Material Sec.6.

LLM-Aided Tag Parsing. While conventional tag parsing methods in vision field [42, 85, 86] often utilize the Natural Language Toolkit (NLTK) [7], this approach struggles with the verbosity of modern caption generators. The captions from *Qwen3-Omni-Captioner*, averaging 388.43 words (see example in Supplementary Material Sec.7.1), are poorly served by NLTK’s reliance on Part-of-Speech

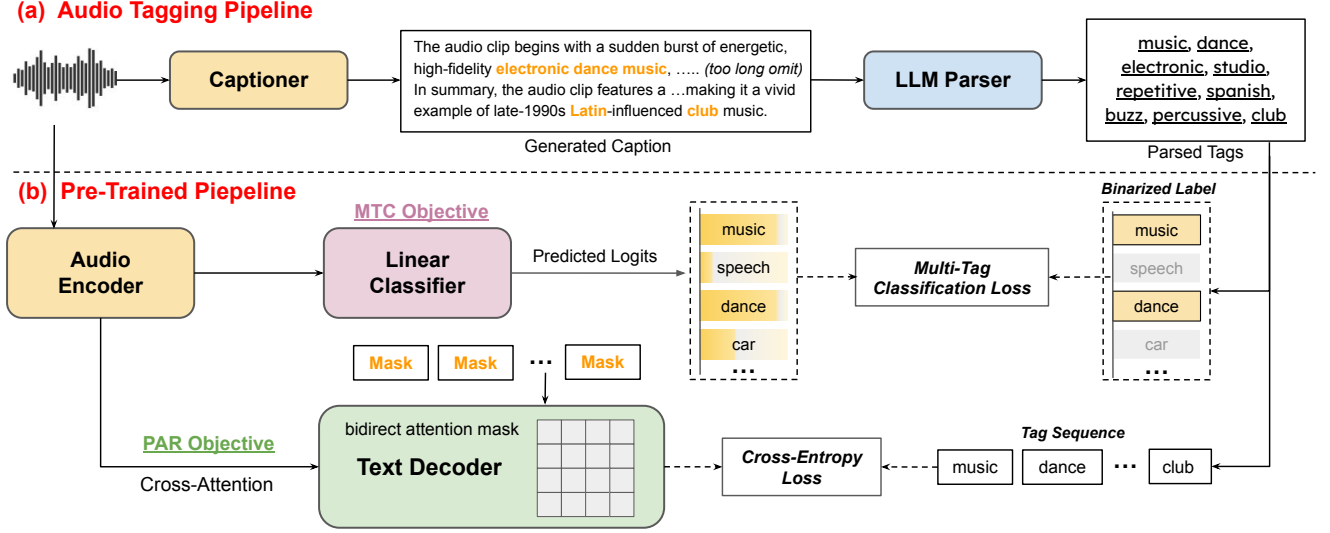


Figure 1. An overview of our method. (a) Audio Tagging Pipeline: we generate tags by processing raw audio through *Qwen3-Omni-Captioner* [87] and an LLM Parser [90]. (b) Pre-Training Pipeline: the audio encoder is then pre-trained using these tags with two supervision objectives: a discriminative multi-tag classification (MTC) Objective (using a linear classifier) and a generative parallel decoding (PAR) Objective (using a text decoder with a bidirectional mask).

(POS) tagging, which hinders a full semantic comprehension. To address this gap, we turn to Large Language Models (LLMs), which are inherently better suited for this challenge due to their advanced capacity for interpreting complex instructions. Inspired by [53] that also employ LLMs for parsing image captions, we adopt *Qwen2.5-7B-Instruct* [90] as our tag parser, g_{parse} . This parser analyzes and distills a set of candidate tags $T'_i = g_{\text{parse}}(c_i)$ from each caption. The specific prompt used to instruct the LLM is detailed in Supplementary Material Sec. 7.2.

Tag Category System Construction. The parsing process generates an audio-tag pair set $\{(a_i, T'_i)\}_{i=1}^N$. We observed that the complete, aggregated tag vocabulary $\mathcal{T}_{\text{pool}} = \bigcup_{i=1}^N T'_i$ was too vast, creating a data sparsity problem that would hinder a multi-tag classification framework. Consequently, we chose to establish fixed-size tag systems, exploring different granularities where the vocabulary size $K \in \{800, 1k, 1.5k, 2k, 3k\}$. The selection of these K tags was guided by the TF-IDF algorithm [73], as it provides a robust measure of a tag’s importance by considering both its local frequency and its global distinctiveness. The score $s(t)$ for a tag $t \in \mathcal{T}_{\text{pool}}$ is:

$$s(t) = df(t) \times \log \left(\frac{N+1}{df(t)+1} \right) \quad (1)$$

where $df(t)$ is the document frequency of tag t . We select the top- K scoring tags to form our final Unified Tag System (UTS), $\mathcal{V} = \{v_1, \dots, v_K\}$. This UTS is novel in that it

unifies speech, music, and environmental sounds. This provides a crucial label foundation, making it possible to develop a single pre-training framework for encoders across all three audio modalities. Further analysis of the unified tag system is available in Sec. 4.4.

3.2. Tag-Oriented Pre-Training

Motivation. Dominant audio-language objectives, such as CLAP [24, 25, 83, 84] or captioning [31, 39], became prevalent because they cleverly use free-form text as a universal supervisory signal, bypassing the need for a unified label set across diverse data. However, their effectiveness has been limited by the quality of their data source; the free-text itself, often scraped from the human annotation [23, 46] or generated by LLMs [59], is typically noisy, sparse, and provides only weak supervisory signals. Our approach begins by solving this fundamental data problem. We derive our Unified Tag System (UTS) from captions generated by a high-fidelity audio captioner (as detailed in Sec. 3.1). This process distills semantically rich, dense, and accurate information into a discrete tag set, creating a strong supervision source. With this powerful UTS in hand, we can now apply inherently strong supervision objectives to pre-train our encoder. We explore two such methods: (1) Multi-Tag Classification (MTC) [95], a direct, validated classification approach, and (2) Parallel Decoding [34], a non-autoregressive generative objective. We now detail these two approaches.

Multi-Tag Classification Objective. With the UTS $\mathcal{V}$ defined, we generate a K -dimensional, multi-hot binary label vector $y_i \in \{0, 1\}^K$ for each audio sample a_i . The j -th element of y_i is defined as:

$$y_{ij} = \begin{cases} 1 & \text{if } v_j \in T'_i \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Samples where y_i becomes a zero-vector are filtered out. This results in our final pre-training dataset $\mathcal{D} = \{(a_i, y_i)\}_{i=1}^M$, where $M \leq N$.

We formulate the pre-training task as a standard multi-label classification problem. Our model, composed of an audio encoder h_θ and a linear classification head W_c , is trained to predict the probability $\hat{y}_i \in [0, 1]^K$ for all K tags in our Unified Tag System. The entire model is trained end-to-end to minimize the Binary Cross-Entropy (BCE) loss $\mathcal{L}_{\text{MTC}}$. This objective compares the model’s predictions $\hat{y}_i$ against the ground-truth multi-hot label vector y_i for each sample a_i , and sum over all K classes, averaged over all M samples in the dataset $\mathcal{D}$:

$$\mathcal{L}_{\text{MTC}} = -\frac{1}{M} \sum_{i=1}^M \sum_{j=1}^K [y_{ij} \log(\hat{y}_{ij}) + (1 - y_{ij}) \log(1 - \hat{y}_{ij})] \quad (3)$$

After pre-training, the classification head W_c is discarded. The learned encoder h_{θ^*} serves as our general-purpose audio representation model, which is then evaluated on a variety of downstream audio understanding tasks.

Generative Pre-training with Parallel Decoding. As an alternative to the MTC objective, we explore a generative pre-training approach that treats the tags as a structured text sequence. This method leverages an encoder-decoder architecture (h_θ and g_ϕ) to explicitly model the mapping from audio to a textual representation of its tags. First, for each audio clip a_i , we convert its ground-truth multi-hot vector y_i back into a canonical text sequence Y_i . We construct this sequence by concatenating all positive tags from our Unified Tag System $\mathcal{V}$ using a comma separator (e.g., $Y_i = \text{"tag_a, tag_d, tag_k"}$). This sequence $Y_i = (y_1, \dots, y_T)$ becomes the target for a generative loss.

A standard autoregressive (AR) captioning objective [88] (Eq. 4) conditions the prediction of the next token y_t on both the audio $z_i^a = h_\theta(a_i)$ and the previously generated tokens $y_{<t}$. However, this autoregressive assumption is a poor fit for our task. The target Y_i is a set of co-occurring tags where the order is arbitrary and carries no contextual logic or linguistic structure. Training an AR model on this data would force it to learn spurious sequential dependencies (e.g., that “bark” must always follow “dog”), which is

both inefficient and incorrect.

$$\mathcal{L}_{\text{ar}} = -\sum_{t=1}^T \log p_\phi(y_t | y_{<t}, z_i^a) \quad (4)$$

To mitigate this, we adopt a Parallel Decoding objective, inspired by non-autoregressive models [34]. This approach is perfectly suited to our set-prediction task. In this mode, we mask the decoder’s input sequence and remove the causal attention mask. This forces the model to predict all T tokens simultaneously, conditioned solely on the audio features:

$$\mathcal{L}_{\text{par}} = -\sum_{t=1}^T \log p_\phi(y_t | z_i^a) \quad (5)$$

This non-autoregressive objective compels the audio encoder h_θ to produce a much richer and more complete representation z_i^a , as this single vector must contain all information necessary to generate the entire tag sequence at once. Our model is trained purely on this $\mathcal{L}_{\text{par}}$ objective, enforcing the strongest possible dependency on the audio encoder and forcing it to serve as the sole source of information for the tag generation task.

3.3. Audio-Language Pre-Training

As an orthogonal line of inquiry, we now investigate the value of the full semantic captions generated by *Qwen3-Omni-Captioner*. Our hypothesis is that the high information quality of these new captions can directly boost the performance of standard audio-language objectives. We utilize the final summary paragraph of the generated caption as the text supervision Y_i , as the full-length captions are too long.

Contrastive Objective. Contrastive pre-training has proven to be a highly effective and scalable method for learning joint audio-language representations [24, 84]. This objective trains two separate encoders, an audio encoder h_θ and a text encoder h_ψ , to project their respective outputs into a shared latent space. The encoders are optimized to maximize the similarity (e.g., cosine) between corresponding (audio, text) pairs (z_i^a, z_i^t) while simultaneously minimizing it for all other mismatched pairs in the batch. This is achieved using a symmetric InfoNCE loss [80]:

$$\mathcal{L}_{\text{con}} = -\frac{1}{2M} \sum_{i=1}^M \left[\log \frac{\exp(\text{sim}(z_i^a, z_i^t)/\tau)}{\sum_{j=1}^M \exp(\text{sim}(z_i^a, z_j^t)/\tau)} + \log \frac{\exp(\text{sim}(z_i^t, z_i^a)/\tau)}{\sum_{j=1}^M \exp(\text{sim}(z_i^t, z_j^a)/\tau)} \right] \quad (6)$$

Captioning Objective. We also explore a generative pre-training objective. As discussed in Sec. 3.2, autoregressive (AR) and parallel (PAR) decoding offer different trade-offs. Here, we adopt the mixed-training strategy from

CapPA [77]. For each minibatch, a fraction of samples is trained with the standard autoregressive objective $\mathcal{L}_{\text{ar}}$ (Eq. 4), while the remainder is trained with the non-autoregressive parallel objective $\mathcal{L}_{\text{par}}$ (Eq. 5). The full generative loss $\mathcal{L}_{\text{gen}}$ is the sum of these two components.

Multi-task Objective. We investigate whether the strong, discrete signal from our MTC objective can be beneficially combined with the dense, generative signal from captioning. We propose a multi-task learning (MTL) framework that jointly optimizes our two strongest objectives: the Multi-Tag Classification objective $\mathcal{L}_{\text{MTC}}$ (from Sec. 3.2) and the Generative Captioning objective $\mathcal{L}_{\text{gen}}$. The final loss is a weighted sum of the two, balanced by a hyperparameter λ :

$$\mathcal{L}_{\text{MTL}} = \mathcal{L}_{\text{MTC}} + \lambda \mathcal{L}_{\text{gen}} \quad (7)$$

This MTL objective forces the audio encoder h_θ to learn representations that are simultaneously discriminative (for MTC) and descriptive (for captioning).

4. Experiment

4.1. Training Setup

Data and Pre-processing. All pre-training experiments are conducted on the audio data from *CaptionStew* 400K-subset. Our tag-oriented objectives (MTC, PAR) are trained on the curated tag labels, while the audio-language objectives (CLAP, Captioning) utilize the final summary paragraph of the generated captions. We process all audio signals by resampling them to 16 kHz and extracting 80-dimensional log-Mel filterbank features, computed using a 25 ms window and a 10 ms hop size. All corresponding text is tokenized using a 50k-vocabulary BPE tokenizer [49].

Model Architectures. Our audio encoder h_θ is the Zipformer-M architecture [92], a model we selected for its high efficiency and strong performance on long sequences. Its design features a U-Net structure with six encoder blocks, which processes features at multiple resolutions to capture both fine- and coarse-grained temporal information. Although originally developed for ASR, recent experiments [2] have confirmed its efficacy as a highly competitive backbone for general audio classification. The text-side architectures vary by objective. For contrastive pre-training, we employ a standard BERT-base text encoder with 12 layers and a 768-dimensional hidden state [20]. For captioning objectives, we use a BART-base decoder with 6 layers and a 768-dimensional hidden state [49]. We selected audio encoder with roughly twice the layer number of the text decoder to ensure comparable training throughput across objectives.

Configuration. All models were trained from scratch on 8 Tesla V100 GPUs, using a per-GPU batch size of 640 audio seconds. We tuned several key hyperparameters for our objectives. For MTC models, we experimented with five tag system vocabularies, $K \in \{800, 1k, 1.5k, 2k, 3k\}$. For the mixed Captioning objective, we used an autoregressive-to-parallel ratio of 0.25. For the Multi-task objective, we tested λ values of 0.2 and 1.0. Training schedules varied by method. The MTC models were trained for 700k steps (approximately 12 days). All other models including Tag Parallel Decoding, CLAP, Captioning, and Multi-task were trained for 400k steps.

4.2. Evaluation Setup

We follow the comprehensive evaluation protocol established in [2] to assess our pre-trained encoders. This protocol assesses models across three distinct axes, with all experiments probing representations from the encoder’s final layer. To ensure a fair comparison, the audio encoder weights are kept **frozen** in all three protocols.

- **Linear Probing.** To assess discriminative capabilities, we train simple linear classifiers for a diverse suite of audio tasks (event classification, sound detection, speaker, and music). We experiment with two pooling mechanisms: mean-pooling and multi-head attention pooling.
- **Audio-language Alignment.** We evaluate both retrieval and captioning capabilities. The retrieval setup pairs the audio encoder with a pretrained RoBERTa-base text encoder [54], whereas the captioning setup utilizes a pretrained BART-base decoder [49] and only finetune cross-attention layers while keeping audio encoders frozen.
- **Open-formed Question Answering.** We benchmark general audio understanding by connecting the frozen encoder to an LLM (*Qwen2.5-7B-Instruct* [90]) and training only a lightweight adapter on the target audio QA datasets and evaluate on corresponding track in AIR-Bench [91].

4.3. Comparative Methods

Following the principles of broad performance benchmarking established in frameworks like SUPERB [81] and HEAR [79], we first establish two primary baseline sets. To benchmark our tag-oriented pre-training methods, we compare against the standard strong supervision paradigm: our Zipformer-M architecture trained on the AudioSet [30]. For our audio-language pre-training methods, we compare against the *Contrastive-scratch* and *Captioning-scratch* trained from scratch on 400k-subset with results reported in [2]. This is a critical comparison, as those models were trained on the exact same 400k audio subset but used original captions, providing a controlled experiment to isolate the impact of caption quality on pre-training performance.

For a broader external comparison, we also selected representative, publicly available SSL models. These baselines

Table 2. Main results on linear probing (with mean-pooling) and audio-language alignment. The linear probing tasks span general audio (FSD-50k [28], VggSound [10], AudioSet-Strong [38]), speech (VoxCeleb2 [16], CREMA-D [9]), and music (MagnaTagATune [82], NSynth [26]). See Supplementary Material Sec. 8 for full details of evaluation datasets and metrics. For audio-language alignment, AC, PSC, and MC refer to the AudioCaps [45], ParaSpeechCaps [22], and MusicCaps [1], respectively. We **bold** the best score among our proposed models. Baseline scores that are underlined are surpassed by their corresponding “Ours” counterpart. † denotes scores quoted from prior work with a consistent evaluation setup.

Methods	Label	Linear Probing							Audio-language Alignment					
		FSD-50k	VggSound	AS-Strong	VoxCeleb2	CREMA-D	MTAT	NSynth	Captioning			Retrieval		
									AC	PSC	MC	AC	PSC	MC
SSL Models														
BEATs [13]	-	0.565 [†]	-	0.034 [†]	-	-	0.400 [†]	75.90 [†]	-	-	-	-	-	-
Wav2vec 2.0 [4]	-	0.342 [†]	-	-	51.60	56.10	0.317 [†]	40.20 [†]	-	-	-	-	-	-
MERT [52]	-	-	-	-	-	-	0.402 [†]	72.60 [†]	-	-	-	-	-	-
Baselines														
MTC (AudioSet)	Tag	0.656	56.46	0.216	<u>18.84</u>	<u>67.14</u>	0.407	67.19	46.67	<u>45.54</u>	<u>22.91</u>	40.46	<u>49.2</u>	24.6
Contrastive-scratch [2]	Caption	0.493	43.78	<u>0.095</u>	38.63	<u>63.74</u>	<u>0.384</u>	<u>60.91</u>	<u>44.50</u>	45.92	<u>22.07</u>	<u>28.73</u>	<u>55.0</u>	<u>19.0</u>
Captioning-scratch [2]	Caption	<u>0.430</u>	<u>39.52</u>	<u>0.077</u>	<u>21.95</u>	<u>60.91</u>	0.378	57.08	<u>43.58</u>	<u>42.85</u>	<u>22.62</u>	<u>26.03</u>	<u>49.2</u>	<u>14.2</u>
Our Tag-Oriented Pre-Trained Models														
MTC (Ours-UTS)	Tag*	0.459	37.70	0.113	37.10	66.01	0.375	63.62	44.40	45.98	23.28	26.14	49.6	15.6
PAR (Ours-UTS)	Tag Sequence*	0.433	39.59	0.121	38.78	62.47	0.381	57.91	44.80	45.66	23.33	26.76	49.8	12.2
Our Audio-Language Pre-Trained Models														
Contrastive-scratch (Ours)	Caption*	0.445	40.78	0.105	33.88	67.29	0.396	61.40	44.54	45.73	22.83	29.66	55.3	19.8
Captioning-scratch (Ours)	Caption*	0.439	39.78	0.087	29.87	64.74	0.377	54.25	45.07	45.81	22.83	26.24	50.2	14.4
Multi-Task (Ours)	Tag*, Caption*	0.485	40.81	0.140	34.62	65.31	0.396	59.94	45.17	46.01	23.09	27.07	49.0	15.8

50k, VggSound), the MTC (AudioSet) model, trained on 5x more data (2M clips), predictably sets a high bar (e.g., Table. 2, FSD-50k: 0.656). However, our models demonstrate dramatic and superior out-of-domain (OOD) generalization. This is most evident in speech tasks. On VoxCeleb2 linear probing, PAR (Ours-UTS) achieves 60.97 (Table. 4) and MTC (Ours-UTS) achieves 37.10 (Table. 2), significantly outperforming the MTC (AudioSet) baseline’s 58.76 and 18.84, respectively. We also observe strong performance from our tag-oriented models in the audio-language alignment benchmarks. On domain-specific tasks, such as music captioning on MusicCaps, our PAR (Ours-UTS) model achieves a T2A Recall@1 of 23.33, outperforming the MTC (AudioSet) baseline of 22.91. This trend is most pronounced in complex reasoning: in Table. 3, MTC (Ours-UTS) achieves near-perfect accuracies on several speech tasks (e.g., 92.64 on gender, 86.59 on age) and wins on Music QA (6.16), decisively crushing the MTC (AudioSet) baseline (5.61). This strongly suggests that the high quality and unified coverage of UTS provides a more potent supervisory signal than a 5x larger dataset.

Audio-Language Pre-Training vs. Baselines. We then evaluate our audio-language pre-training models, which are trained on our SOTA-generated captions (Caption*), against the [2] baselines, which were trained on the exact same 400k audio data but used low-quality and information-sparse captions. Our models demonstrate a clear and consistent advantage, surpassing the [2] baselines on the majority of evaluation tasks, as indicated by the underlined metrics across our tables. This trend is especially pronounced in Table. 4), where Contrastive-scratch (Ours) outperforms the baseline on 4 out of 6 linear probing tasks. Furthermore, in the audio-language alignment benchmarks

(Table. 2), our Captioning-scratch (Ours) model outperforms the Captioning-scratch [2] baseline across all evaluated metrics. This comprehensive set of results strongly indicates that the higher information quality and semantic density of our Caption* source directly translate into more robust and versatile encoders compared to models trained on original captions.

Table 3. Open-formed Question Answering (QA) Results on the AIR-Bench [91]. The *Sound* and *Music* columns report 10-point scores from open-ended questions, evaluated by GPT-4o [63]. The five scores in the *Speech* column correspond to the classification accuracy on the Emotion-MELD, Emotion-IEMOCAP, Gender-MELD, Gender-common, and Age tasks, respectively.

Methods	Open-formed QA			
	Sound	Speech (emotion/gender/age)		Music
<i>Baselines</i>				
MTC (AudioSet)	7.01	47.16/25.90/47.27	45.06/37.24	<u>5.61</u>
Contrastive-scratch [2]	<u>5.69</u>	29.61/48.79/55.25	84.85/86.59	<u>5.99</u>
Captioning-scratch [2]	<u>6.25</u>	<u>22.93</u> /37.40/70.20	78.93/40.14	<u>5.73</u>
<i>Our Tag-Oriented Pre-Trained Models</i>				
MTC (Ours-UTS)	6.68	39.19/48.79/74.45	92.64/86.59	6.16
PAR (Ours-UTS)	6.59	17.90/38.64/47.82	37.40/57.16	6.03
<i>Our Audio-Language Pre-Trained Models</i>				
Contrastive-scratch (Ours)	5.78	44.08 /41.34/73.11	77.82/40.48	6.02
Captioning-scratch (Ours)	6.35	23.47/29.90/46.40	28.02/18.82	5.93
Multi-Task (Ours)	6.60	41.70/32.86/53.38	57.00/65.47	6.15

Analysis of Different Pre-Training Objectives. Our results reveal a clear specialization of objectives across the three evaluation protocols. No single method excels at all tasks; instead, the pre-training objective creates a strong inductive bias. For Linear Probing, the choice of pooling mechanism is critical. While our Multi-Task model shows the broadest strength in mean-pooling (Table. 2), the Contrastive-scratch (Ours) model becomes dominant when using multi-head attention pooling (Table. 4), sweeping all 6 tasks. This suggests its learned features are uniquely

Methods	Linear Probing					
	FSD-50k	VggSound	VoxCeleb2	CREMA-D	MTAT	NSynth
Baselines						
MTC (AudioSet)	0.656	56.23	<u>58.76</u>	72.52	0.405	74.80
Contrastive-scratch [2]	0.534	46.79	<u>70.18</u>	<u>69.97</u>	<u>0.385</u>	<u>70.19</u>
Captioning-scratch [2]	<u>0.483</u>	43.43	<u>44.51</u>	<u>66.29</u>	<u>0.382</u>	<u>68.31</u>
Our Tag-Oriented Pre-Trained Models						
MTC (Ours-UTS)	0.478	40.02	52.52	70.97	0.376	69.60
PAR (Ours-UTS)	0.489	43.27	60.97	69.98	0.388	68.97
Our Audio-Language Pre-Trained Models						
Contrastive-scratch (Ours)	0.514	45.63	71.63	72.39	0.401	71.22
Captioning-scratch (Ours)	0.485	43.53	49.94	68.56	0.386	67.29
Multi-Task (Ours)	0.503	43.35	53.40	70.69	0.399	68.95

Table 4. Linear probing results with multi-head attention pooling.

suiting to attention-based aggregation. For Audio-Language Alignment, Contrastive-scratch (Ours) expectedly wins in retrieval. However, we observe that the Captioning-scratch (Ours) model is not the best performer on captioning tasks; this honor goes to the Multi-Task (Ours) model, suggesting that the strong MTC signal acts as a powerful regularizer for the generative objective. Finally, for Open-formed QA, MTC (Ours-UTS) is the clear champion (Table. 3), demonstrating that its fine-grained, strongly-supervised tag-based learning is most effective for instilling the factual, discriminative knowledge required for LLM reasoning. Conversely, the PAR model, while strong in probing, fails catastrophically in QA, highlighting its extreme task specialization.

Given these clear trade-offs, the Multi-Task (Ours) model emerges as a powerful and balanced hybrid. It achieves the best-in-class performance among all our models on general-purpose probing tasks (e.g., FSD-50k, VggSound in Table 2), outperforming other objectives on captioning tasks (AudioCaps, ParaSpeechCaps), and delivering strong results on Music QA (Table 3: 6.15). This demonstrates that combining the discrete, strong signal of our Tag* with the descriptive, rich signal of our Caption* yields a highly effective “jack-of-all-trades” representation.

4.6. Discussion

The Impact of Tag System Size. We analyze the effect of our UTS vocabulary size K , as shown in Fig. 3. We plotted the normalized average performance of our MTC models ($K = 800$ to $3k$) across three major task categories (linear probing, captioning and retrieval). For linear probing performance, we only use the results from mean-pooling, as it better reflects the intrinsic quality of the learned representations. These scores are computed by first setting the MTC (AudioSet) baseline as 100% for each individual task, and then averaging the relative percentage scores of our models within that category. For linear probing and retrieval, performance clearly peaks at $K=1.5k$ or $K=2k$, after which it begins to plateau or decline. This suggests a trade-off: a larger K provides a more granular vocabulary but also introduces data sparsity. For our 400k dataset, a vocabulary size of 1.5k-2k appears to be the “sweet spot” that bal-

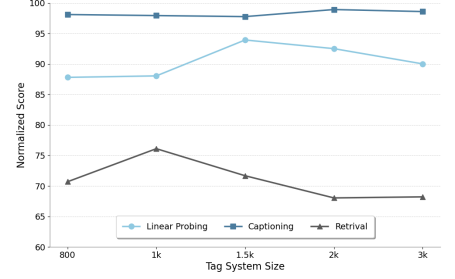


Figure 3. Impact of Tag System Size

ances semantic richness and trainability. Generative tasks like Captioning appear more robust to larger K values.

Data Quality as the Primary Driver of Performance.

Our comprehensive results suggest that the quality of the supervision source is a more critical determinant of performance than the specific pre-training objective. We observe that on our 400k dataset, the performance gap between our tag-oriented and audio-language models is often smaller than their collective gap against the baselines. This “data-first” principle [71, 75] is evident in two key comparisons. First, our tag-oriented models achieve superior out-of-domain generalization compared to the MTC (AudioSet) baseline despite using a 5x smaller dataset; this demonstrates that the unified coverage of our UTS is more impactful than raw data volume. Second, when controlling for data size, our audio-language models (trained on SOTA-captions) consistently outperform the baselines (trained on original captions), proving that label richness directly translates to more robust encoders. This leads to our central observation: data matters most. The choice of pre-training objective appears to be a secondary factor, primarily leading to different task specializations. This provides a clear lesson for the audio pre-training community. While objective design is important, future efforts should be directed toward the supervision source itself, either by developing even larger, more comprehensive UTS or by advancing high-fidelity captioners, as rich text is itself the ultimate form of universal audio supervision. Fundamentally, this aligns with the core goal of learning: ensuring the data and its supervisory label are more closely and richly matched.

5. Limitation and Conclusion

We proposed a new data-centric paradigm for general-purpose audio pre-training, and introduced a large-scale Unified Tag System (UTS), the first of its kind to bridge speech, music, and environmental sounds. Our systematic study demonstrates that models trained on this “strong source” outperform baselines on larger, non-unified datasets. This confirms that data quality is the primary driver of performance, while the choice of objective dictates

downstream task specialization. We also acknowledge limitations. Our UTS is inherently biased by its single “teacher” captioner, and the interplay between data quality and volume at massive scales remains an open question. This observed task specialization also highlights a key challenge for future work: designing a single, unified objective that can excel across all downstream tasks without compromise.

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Unlocking Strong Supervision: A Data-Centric Study of General-Purpose Audio Pre-Training Methods

Supplementary Material

6. Audio Source of Training Data

We utilize 400k audio subset from *CaptionStew* [2], a composite dataset aggregating 8 open-source collections to mitigate data scarcity and enhance diversity in audio pre-training. As outlined in Table. 5, the data spans diverse acoustic domains, including environmental sounds, music, and expressive speech.

7. Label Construction

As described in main text, our label construction pipeline comprises two distinct stages: audio caption generation and LLM-aided tag parsing. In this section, we provide comprehensive implementation details and hyperparameter configurations for both steps.

7.1. Captioner Details and Examples

We employ Qwen3-Omni-30B-A3B-Captioner [87] for audio caption generation, utilizing the vLLM engine with 8-way tensor parallelism and bfloat16 precision to ensure efficiency. The data loading pipeline is managed by Lhotse [98], which applies dynamic bucketing with a batch size of 32 to optimize GPU utilization. For generation, we use nucleus sampling with a temperature of 0.6 and top- p of 0.95 to balance diversity and coherence. We provide samples of the generated captions and corresponding audio files in the supplementary material.

7.2. LLM Parser

We employ Qwen2.5-7B-Instruct [90] as a semantic parser to extract structured tags from unstructured audio captions. Utilizing vLLM with 4-way tensor parallelism, we prompt the model to generate 5–10 concise, one-word labels per sample in JSON format. The instruction ensures coverage of key acoustic dimensions—including scene, sound source, and event type—using sampling parameters of $T = 0.3$ and $p = 0.9$. We provided the full text of the instruction prompt.

8. Evaluation Details

Table. 6 provides a comprehensive breakdown of the datasets and metrics used for evaluate audio representation quality. The assessment covers three distinct protocols: linear probing, audio-language alignment and open-form question answering.

9. Complete Results

We have the complete results on three evaluation axes in Table. 7 and Table. 8. The main additions are the results for five values of K ranging from 800 to 3k, as well as the results for the multi-task setting where λ is set to 0.25 and 1, respectively.

Table 5. Overview of the public datasets constituting CaptionStew. The table summarizes their scale, domain coverage, audio sources, and diverse captioning pipelines (from human annotation to LLM generation).

Dataset	#audio/#cap	Domain	Audio source	Caption style	Caption generation pipeline
AudioCaps	46k/46k	general (environmental, human/animal sounds)	AudioSet	Human-annotated, short description	crowdsourced
Clotho	5k/25k	environmental sounds	FreeSound	Human-annotated, short description	crowdsourced
MusicCaps	3k/3k	music	AudioSet	Expert musician-written, multi-sentence, fine-grained description	expert curation
WavCaps	400k/400k	general (environmental, human/animal sounds)	AudioSet BBC Sound Effect FreeSound SoundBible	LLM-refined captions	three-stage pipeline: web-crawled raw descriptions → ChatGPT rewrite → filtering
AudioSetCaps	1.9M/1.9M 4.0M/4.0M 182k/182k	general (environmental, human/animal sounds)	AudioSet YouTube8M VggSound	LLM-generated, detailed, multi-sentence description	three-stage pipeline: LALM attribute extraction → LLM captioning → CLAP-based filtering
FusionAudio	1.2M/1.2M	general (environmental, human/animal sounds)	AudioSet	LLM-augmented, multi-sentence, visual-enhanced description	multimodal context fusion (audio, visual, metadata) + LLM captioning
JamendoMaxCap	360k/1.8M	music	Jamendo Platform	LLM-augmented, multi-sentence, fine-grained music description	retrieval-based metadata imputation + LLM captioning
ParaSpeechCaps	116k/116k (base) 924k/924k (scaled)	expressive speech	VoxCeleb1 VoxCeleb2 EARS Expresso Emilia	Human-annotated/LLM-augmented, speaking-style description	crowdsourced / retrieval-based metadata imputation + LALM captioning

Prompt :

You are a labeling assistant for audio descriptions. Extract essential, compact English labels for the audio caption.

Output rules:

- Output JSON only with a single key "labels": a list of 10 labels.
- Each label must be ONE WORD.
- Return ONLY a raw JSON object with key "labels". Do not use code fences or any extra text.
- Use lowercase.
- No full sentences, no punctuation beyond hyphens.
- Prefer canonical, domain-relevant terms with high reusability across captions.
- Avoid synonyms: collapse to a single consistent term set (e.g., always use "car" not "automobile").
- Avoid rare or obscure words; choose common, widely recognized terms.
- Cover five aspects when possible: scene/environment, sound sources, sound types/events, audio/production traits, semantic function/intent.
- Avoid duplicates.

Expected JSON:

"labels": ["xxx", "xxx", "xxx", "xxx", "xxx"]

Now process the following caption and return JSON only:

CAPTION:
{caption}

Table 6. Overview of the evaluation datasets used for assessing audio representation quality. [†]Evaluated by GPT-4o following the AIR-Bench protocol. [‡]Synthesized from publicly available speech datasets [3, 8, 9, 55, 64] using fixed question templates.

Evaluation Dataset	Task	#samples	#class	train	eval	Metrics
FSD-50k	Multi-label audio event classification	37,168 / 10,231	200	✓	✓	mAP
VggSound	Single-label audio event classification	183,730 / 15,446	309	✓	✓	accuracy
VoxCeleb2	Speaker identification	1,092,009 / 36,693	5,994	✓	✓	accuracy
CREMA-D	Speech emotion recognition	6,030 / 706	6	✓	✓	accuracy
MagnaTagATune	Music tagging	19,425 / 4,856	50	✓	✓	mAP
NSynth	Musical instrument classification	289,205 / 4,096	11	✓	✓	accuracy
AudioSet-strong	Sound event detection	103,463 / 16,996	456	✓	✓	PSDS1
AudioCaps	Text-to-audio retrieval	49,838 / 975	–	✓	✓	Recall@1
ParaSpeechCaps	Audio captioning	116,516 / 500	–	✓	✓	RougeL
MusicCaps		2,663 / 500	–	✓	✓	
ClothoAQA		7,044	–	✓	×	
In-house SpeechQA [‡]		160,000	–	✓	×	
MusicQA		70,011	–	✓	×	
AIRBench-chat-sound	Open-formed question answering	400	–	×	✓	Score [†]
AIRBench-foundation-emotion		1,000	–	×	✓	
AIRBench-foundation-gender		1,000	–	×	✓	
AIRBench-foundation-age		1,000	–	×	✓	

Table 7. **Complete** results on linear probing (with mean-pooling) and audio-language alignment. The linear probing tasks span general audio (FSD-50k [28], VggSound [10], AudioSet-Strong [38]), speech (VoxCeleb2 [16], CREMA-D [9]), and music (MagnaTagATune [82], NSynth [26]). For audio-language alignment, AC, PSC, and MC refer to the AudioCaps [45], ParaSpeechCaps [22], and MusicCaps [1], respectively. We **bold** the best score in our tag-oriented models and audio-language models respectively. Baseline scores that are underlined are surpassed by their corresponding “Ours” counterpart. [†] denotes scores quoted from prior work with a consistent evaluation setup.

Methods	Label	Linear Probing							Audio-language Alignment					
		FSD-50k	VggSound	AS-Strong	VoxCeleb2	CREMA-D	MTAT	NSynth	Captioning			Retrieval		
SSL Models														
BEATs [13]	-	0.565 [†]	-	0.034 [†]	-	-	0.400 [†]	75.90 [†]	-	-	-	-	-	-
Wav2vec 2.0 [4]	-	0.342 [†]	-	-	51.60	56.10	0.317 [†]	40.20 [†]	-	-	-	-	-	-
MERT [52]	-	-	-	-	-	-	0.402 [†]	72.60 [†]	-	-	-	-	-	-
Baselines														
MTC (AudioSet)	Tag	0.656	56.46	0.216	<u>18.84</u>	<u>67.14</u>	0.407	67.19	46.67	<u>45.54</u>	<u>22.91</u>	40.46	<u>49.2</u>	24.6
Contrastive-scratch [2]	Caption	0.493	43.78	<u>0.095</u>	38.63	<u>63.74</u>	<u>0.384</u>	<u>60.91</u>	<u>44.50</u>	45.92	<u>22.07</u>	<u>28.73</u>	<u>55.0</u>	<u>19.0</u>
Captioning-scratch [2]	Caption	<u>0.430</u>	<u>39.52</u>	<u>0.077</u>	<u>21.95</u>	<u>60.91</u>	0.378	57.08	<u>43.58</u>	<u>42.85</u>	<u>22.62</u>	<u>26.03</u>	<u>49.2</u>	<u>14.2</u>
Our Tag-Oriented Pre-Trained Models														
MTC (Ours-UTS, K=800)	Tag*	0.448	37.01	0.095	30.04	61.34	0.375	62.84	44.40	45.72	22.62	23.96	48.4	13.4
MTC (Ours-UTS, K=1k)	Tag*	0.455	36.79	0.095	30.26	62.90	0.373	61.50	44.20	45.98	22.47	25.93	49.6	15.6
MTC (Ours-UTS, K=1.5k)	Tag*	0.459	37.70	0.104	37.10	64.31	0.368	60.01	43.66	45.50	22.86	26.41	48.4	12.8
MTC (Ours-UTS, K=2k)	Tag*	0.450	37.48	0.113	33.22	66.01	0.370	63.62	44.06	45.86	23.28	24.69	46.8	11.8
MTC (Ours-UTS, K=3k)	Tag*	0.449	37.68	0.113	30.63	65.02	0.373	61.57	44.09	45.87	23.03	24.90	46.0	12.2
PAR (Ours-UTS)	Tag Sequence*	0.433	39.59	0.121	38.78	62.47	0.381	57.91	44.80	45.66	23.33	26.76	49.8	12.2
Our Audio-Language Pre-Trained Models														
Contrastive-scratch (Ours)	Caption*	0.445	40.78	0.105	33.88	67.29	0.396	61.40	44.54	45.73	22.83	29.66	55.3	19.8
Captioning-scratch (Ours)	Caption*	0.439	39.78	0.087	29.87	64.74	0.377	54.25	45.07	45.81	22.83	26.24	50.2	14.4
Multi-Task (Ours, $\lambda = 0.2$)	Tag*, Caption*	0.468	39.47	0.130	34.62	62.61	0.386	57.71	44.88	46.01	23.20	25.21	49.0	13.5
Multi-Task (Ours, $\lambda = 1$)	Tag*, Caption*	0.485	40.81	0.140	33.12	65.31	0.396	59.94	45.17	45.55	23.09	27.07	47.0	15.8

Table 8. **Complete** results on linear probing (with multi-head attention pooling) and open-formed QA. The *Sound* and *Music* columns report 10-point scores on open-ended questions from AIR-Bench, evaluated by GPT-4o [63]. The five scores in the *Speech* column correspond to the classification accuracy on the Emotion-MELD, Emotion-IEMOCAP, Gender-MELD, Gender-common, and Age tasks, respectively.

Methods	Linear Probing						Open-formed QA			
	FSD-50k	VggSound	VoxCeleb2	CREMA-D	MTAT	NSynth	Sound	Speech (emotion/gender/age)		Music
Baselines										
MTC (AudioSet)	0.656	56.23	<u>58.76</u>	72.52	0.405	74.80	7.01	47.16/25.90/47.27/45.06/37.24		<u>5.61</u>
Contrastive-scratch [2]	0.534	46.79	<u>70.18</u>	<u>69.97</u>	<u>0.385</u>	<u>70.19</u>	<u>5.69</u>	<u>29.61</u> /48.79/55.25/84.85/86.59		<u>5.99</u>
Captioning-scratch [2]	<u>0.483</u>	43.43	<u>44.51</u>	<u>66.29</u>	<u>0.382</u>	<u>68.31</u>	<u>6.25</u>	22.93/37.40/70.20/78.93/40.14		<u>5.73</u>
Our Tag-Oriented Pre-Trained Models										
MTC (Ours-UTS, $K=800$)	0.469	39.56	45.98	67.57	0.376	68.97	6.54	29.61/ 48.79 /55.25/84.85/ 86.59		5.99
MTC (Ours-UTS, $K=1k$)	0.472	39.84	46.02	68.71	0.376	69.56	6.46	18.45/30.10/48.91/56.35/58.66		5.77
MTC (Ours-UTS, $K=1.5k$)	0.477	39.94	52.52	69.41	0.371	69.60	6.48	39.19 /32.10/62.66/82.06/56.26		5.80
MTC (Ours-UTS, $K=2k$)	0.478	39.78	48.13	70.97	0.371	68.92	6.47	26.20/32.60/54.15/73.49/40.84		6.16
MTC (Ours-UTS, $K=3k$)	0.471	40.02	45.11	69.29	0.371	69.41	6.68	22.38/29.70/ 74.45 / 92.64 /63.06		5.95
PAR (Ours-UTS)	0.489	43.27	60.97	69.98	0.388	68.97	6.59	17.90/38.64/47.82/37.40/57.16		6.03
Our Audio-Language Pre-Trained Models										
Contrastive-scratch (Ours)	0.514	45.63	71.63	72.39	0.401	71.22	5.78	44.08 / 41.34 / 73.11 / 77.82 /40.48		6.02
Captioning-scratch (Ours)	0.485	43.53	49.94	68.56	0.386	67.29	6.35	23.47/29.90/46.40/28.02/18.82		5.93
Multi-Task (Ours, $\lambda = 0.25$)	0.490	41.99	50.80	69.70	0.399	67.33	6.60	41.70/26.90/57.10/53.02/45.15		6.09
Multi-Task (Ours, $\lambda = 1$)	0.503	43.35	53.4	70.69	0.393	68.95	6.38	24.62/32.86/53.38/57.00/ 65.47		6.15

Audio id: Y7IOszZm4n.I

The audio begins in a quiet, indoor environment, marked by a persistent low-frequency hum and a faint high-frequency hiss, likely produced by an appliance or HVAC system. The atmosphere is calm and still, with no background speech or music. A soft, low-pitched rustling sound, characteristic of a dog's movement through dry leaves or grass, is heard near the recording device, followed by a subtle, low-frequency thump suggesting the dog's body settling or shifting weight. Shortly after, a faint, high-pitched squeak—possibly from a toy or collar—adds to the sense of a dog's presence. A sharp, brief rustling indicates the dog's claws or paws moving across a rough surface.

Suddenly, a male voice, calm and authoritative, commands, "Come." His speech is clear, mid-to-low in pitch, and delivered with a neutral General American accent, indicating a confident and focused tone. Immediately following the command, a sharp, percussive tongue click is produced, serving as a non-verbal cue to reinforce the verbal command. The dog responds with a distinct, high-pitched bark, signaling acknowledgment and readiness. The dog's movement continues with a pronounced rustling sound as it shifts or stands up, accompanied by the jingle of metal tags on its collar.

The male then issues the command, "Drop," in a firm and slightly more urgent tone, indicating an expectation of a specific action. The dog's response is immediate and enthusiastic: a rapid burst of barks and excited yips, accompanied by continued rustling and the jingling of tags as it moves energetically, possibly jumping or turning. The male repeats the command, "Drop," with increased urgency, reflecting his expectation for the dog to comply. The dog's barking escalates in intensity and pitch, now sounding frantic and excited, while the rustling and jingling persist. The clip ends abruptly, with all sounds ceasing at once, leaving no fade-out or lingering noise.

Throughout the recording, the audio quality is moderate to low, with clear speech but noticeable hiss and hum, and the environment appears to be a small to medium-sized, enclosed indoor space with hard surfaces. The dog's vocalizations and movements are prominent and close to the microphone, indicating the device is positioned near both the dog and the handler. The interaction is structured and purposeful, with the handler issuing commands and the dog responding with excitement and obedience. The context suggests a focused training or play session between a handler and a small or medium-sized, energetic dog, typical of North American pet ownership and training practices.

In summary, this audio clip documents a succinct, indoor training exchange between a man and his dog. The man gives the commands "Come" and "Drop," reinforced by a tongue click, and the dog responds with barks, yips, rustling, and tag jingling. The setting is a quiet, enclosed space with a steady background hum and hiss, and the interaction exemplifies a structured, purposeful moment of communication and obedience training typical of North American pet culture.

Tags: ["indoors", "appliance", "rustling", "dog", "squeak", "command", "click", "bark", "training", "hiss"]

Audio id: MG0ThC4TSoQ_94_10

The audio clip begins with a burst of energetic, high-fidelity electronic music characteristic of late 1980s to early 1990s Chicago House and Acid House. The soundscape is dominated by a punchy four-on-the-floor kick drum, sharp synthetic claps or snares, and a crisp hi-hat pattern, all tightly sequenced to drive the rhythm. A deep, resonant synth bassline—likely from a Roland TB-303 or similar—delivers the iconic acid squelch, while a bright, syncopated arpeggiated synthesizer line adds melodic complexity. A recurring, high-pitched, and slightly distorted vocal sample of a female voice saying “for you” is chopped and looped, functioning as both a rhythmic and melodic hook. The stereo image is expansive, with drums and bass centered, synth layers panned for width, and vocal samples and percussion spread across the field. The overall production is clean, loud, and compressed, with no ambient noise or room tone, suggesting a studio or digital origin.

At the five-second mark, the intensity peaks as the bass and synth layers momentarily drop out, leaving the drums and a sharp snare-like clap isolated. This creates a brief, suspenseful moment of anticipation before the full instrumental arrangement resumes at full volume. The acid bassline and arpeggiated synth return, and a new vocal sample enters: a processed male voice, heavily treated with reverb and echo, delivers the phrase “if it’s good to you.” The vocal is delivered in a rhythmic, spoken-word style, with the word “good” notably emphasized and repeated, creating a hypnotic effect. The phrase is clipped to fit the musical rhythm and is repeated for emphasis, with no additional lyrics or dialogue.

As the track reaches its conclusion, all elements—drums, bass, synths, and vocals—abruptly cut off, leaving no fade-out or lingering sound. The sudden silence suggests the clip is an excerpt from a longer, continuous musical work.

In summary, this audio is a meticulously produced, high-energy excerpt from a Chicago House/Acid House track, distinguished by its classic drum machine patterns, acid bassline, arpeggiated synth hooks, and iconic vocal samples. The production is studio-clean, with deliberate stereo placement and effects, and the clip serves as a vivid snapshot of early rave culture, evoking nostalgia for the golden age of electronic dance music.

Tags: [“music”, “electronic”, “house”, “acid”, “drum”, “bass”, “synth”, “vocal”, “production”, “nostalgia”]

Audio id: YITcnYeETGUE

The audio clip begins with a faint rustle, likely from the speaker adjusting a microphone or clothing, immediately followed by a soft inhalation, indicating the speaker is preparing to address the audience. The recording is marked by a persistent, low-level hiss from the microphone's noise floor, and a subtle, low-frequency hum typical of a small, untreated indoor space. A single male speaker, with a clear, slightly nasal Italian voice and a regional accent from Northern or Central Italy, speaks in a calm, measured, and friendly manner. He uses a conversational tone, employing filler words and informal constructions, which suggests he is addressing a general audience in a relaxed, personal style.

The speaker's speech starts abruptly, mid-sentence: "...praticamente dicevo un treno Shinkansen l'ottavo giorno, andate verso Hiroshima. Appena arrivati a Hiroshima, e visitate il parco della pace, vicino al quale potete prendere un hotel..." ("...practically, I was saying, a Shinkansen train on the eighth day, go towards Hiroshima. As soon as you arrive in Hiroshima, and visit the Peace Park, near which you can take a hotel..."). The language is Italian with English loanwords ("Shinkansen"), reflecting modern Italian usage and a travel-oriented context. The speaker references the eighth day of a trip, the Shinkansen high-speed train, the city of Hiroshima, and the Peace Park, indicating he is providing travel advice or narrating a personal itinerary. His tone remains neutral and friendly, with no emotional emphasis or change in delivery. The recording ends abruptly, cutting off the final word "hotel" and leaving the speech unfinished, which suggests an accidental or intentional truncation. Throughout the clip, the audio quality is moderate, with clear speech but some muffling due to the acoustics of the small room. The frequency response is limited, lacking deep bass and crisp treble, and the signal is free from clipping or distortion. The speaker's voice is close-mic'd, and there are no other voices, background noises, or music. The content and style indicate that the speaker is likely a young-to-middle-aged Italian male sharing travel advice or narrating a personal journey, aimed at an audience interested in Japanese travel, with a focus on Hiroshima and its Peace Park. The use of Italian with English loanwords and the informal, conversational approach suggest a personal or semi-professional travel vlog or podcast.

In summary, the audio captures a brief, unfinished segment of a single male Italian speaker providing travel advice, specifically mentioning the eighth day of a trip to Hiroshima via Shinkansen, the visit to the Peace Park, and nearby hotel options. The recording is clear and conversational, set in a small indoor space with mild background hiss and hum, and ends abruptly mid-word, reflecting a personal travel narrative intended for a general audience.

Tags: ["speech", "adjustment", "inhale", "hiss", "hum", "italian", "calm", "conversational", "travel", "abrupt"]

Audio id: 963530_0

The audio clip is a 30-second, low-fidelity, stereo field recording featuring a solo acoustic guitar performance. The guitar is captured in a very intimate, close-mic'd manner, revealing not only the instrument's full-bodied sound but also every nuance of the performer's technique: finger slides, string squeaks, and the tactile sounds of the performer adjusting their hand and body.

The performance is structured in three clear, non-repetitive sections. It begins with a slow, bluesy riff in a minor key, played in a fingerstyle manner. The second section introduces a more rhythmic, percussive strumming pattern, while the third section features a melodic, arpeggiated passage. Throughout, the playing is expressive and nuanced, with deliberate dynamics and a sense of improvisation. The musical style, including the choice of chords, fingerstyle technique, and melodic phrasing, is strongly rooted in the American folk-blues tradition, reminiscent of early 20th-century rural or country blues, but presented with modern recording clarity.

The recording environment is outdoors, as evidenced by a constant, low-frequency wind rumble and occasional rustling of clothing, but there are no other environmental or human sounds. The absence of any reverb, echo, or ambient noise points to an open, rural or semi-rural location, likely away from urban or natural sound sources.

The technical quality of the recording is compromised by a persistent hiss and wind rumble, and the guitar's resonance is slightly muffled, indicating the use of a consumer-grade recording device. The stereo field is narrow, with the guitar and ambient sounds centered, and the overall sound lacks the brightness and detail of a professional studio recording.

The clip ends abruptly, with a sharp click indicating the recording was manually stopped. Immediately after, a short, low-frequency electronic tone is heard, possibly a notification or system sound from the recording device, marking the end of the session.

In summary, the audio is a raw, unedited field recording of a solo acoustic guitar performance in a wind-swept outdoor setting, capturing the essence of American folk-blues tradition in a modern, intimate, and slightly lo-fi context. The performer's technical skill and expressive intent are clear, but the recording's limitations and environmental noise contribute to a sense of authenticity and immediacy, evoking a solitary, reflective musical moment in a rural landscape.

Tags: ["outdoor", "guitar", "blues", "intimate", "low_fidelity", "field_recording", "folk", "acoustic", "expressive", "natural_noise"]

Audio id: 71xd4wFkozw_62_10

The audio clip opens with a male commentator, speaking in a clear, urgent, and slightly raised North American accent, his voice reverberating through a large, echoic space. He poses a rhetorical question: “Why would he let his mouth get into the point of swelling up that much?” His delivery is fast-paced and marked by rising intonation, reflecting tension and disbelief at the apparent risk involved in a physical altercation. The speech is accompanied by a persistent, low-level hiss, typical of analog tape recordings from the late 20th century, and is further colored by the room’s acoustics, which impart a hollow echo to the commentator’s voice.

As the commentator speaks, the soundscape intensifies with a series of sharp, percussive impacts—dry, hollow thuds that suggest blows landing on flesh, particularly on the head or face, and are interlaced with guttural grunts and strained vocalizations. These non-verbal sounds, including a forceful grunt around the midpoint, indicate the physical exertion and pain of combatants, likely engaged in a martial arts or boxing match. The impacts and vocalizations reverberate through the same spacious, hard-walled venue, with the microphone capturing both direct and ambient sound.

A female voice soon enters, her tone urgent and high-pitched, calling out “Yeah!” and “Come on!” in quick succession. Her shouts, delivered with the same reverberant qualities as the commentator’s, serve as a rallying cry or encouragement, possibly directed at the fighters or the audience. Her brief interjections are marked by clarity and emotional intensity, cutting through the ongoing commotion.

The recording ends abruptly with the female’s last exclamation, leaving the fight unresolved and the soundscape abruptly silenced, a sign of analog tape’s limitations.

In summary, this analog recording captures a moment of intense physical competition within a large, reverberant venue, likely a gymnasium or arena, during a martial arts or boxing match. The male commentator’s urgent rhetorical question and the female’s supportive shouts frame the ongoing struggle, which is punctuated by realistic impact sounds and strained vocalizations. The technical imperfections of the recording—hiss, echo, and abrupt ending—reinforce its authenticity and place it firmly within the context of late 20th-century North American sporting events.

Tags: [“commentator”, “male”, “urgent”, “hiss”, “impact”, “grunts”, “female”, “encouragement”, “venue”, “boxing”]